

Tai Lin

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Founding engineer who embeds with customers to take frontier-model systems from ambiguous requirements to production. At HighFi, I was the primary technical point of contact scaling our first two customers from \$0 to \$350K ARR, owning discovery, integration, and rollout of multi-step AI agents, configurable financial engines, and the agent infrastructure underneath them in a regulated capital-markets domain.

EXPERIENCE

Founding Software Engineer | HighFi

Mar 2023 – Present

AI-enabled capital markets infrastructure · TypeScript, PostgreSQL, Next.js, Vercel AI SDK, Pulumi, GCP

AI Agent Products (Penny & Millie)

- Shipped Penny, an in-app capital-markets agent (Claude + Gemini, streaming) for real-time risk reporting, funding requests, and ledger management. Ownership from concept to production, replacing a manual funding workflow that previously took 2 hours and 10 steps per request.
- Architected and shipped Millie, a context-aware Slack agent that builds authorized toolsets per request and runs up to 5-step multi-tool execution for funding requests, SOFR updates, and portfolio reporting. This includes webhook verification, event deduplication, and thread-persistent history in PostgreSQL. HighFi's largest enterprise customer used Millie for daily & monthly compliance reporting workflows.

Platform (Agent Harness & Context Layer)

- Designed and built a production AI agent harness with typed DI, structured step tracing, stateful inter-agent hand-offs, and per-step observability. Also doubles as the evaluation surface for catching agent regressions and failures before they reach production. Spawned 6 agents across a multi-tenant capital-markets platform.
- Built a capability registry that gates LLM tool access by tenant configuration and user role, generating dynamic system prompts and workflow manifests for auditable multi-step financial operations.

Core Systems

- Designed a configuration-driven financial calculation engine with dependency graph evaluation and runtime expressions, enabling customer-defined eligibility, concentration, and facility formulas without code changes.
- Architected the lending Assignment Engine, a deterministic allocation system that assigns receivables under borrowing-base constraints, enforces concentration limits, and supports configurable add/remove prioritization strategies.

Agentic Infrastructure

- Built Night Agent, a three-stage Claude Code pipeline on GitHub Actions (Planner → Implementer → QA) with a zero-trust shell harness *below* the prompt layer: every command clears a user-defined allowlist before execution, a structural safety guarantee no prompt-level guardrail can give. Specs by day, PRs reviewed at 8 AM.

Software Engineer → Software Engineer II | Utradea

Mar 2021 – Feb 2023

Social investing platform → B2B API layer · Node.js, MongoDB, PostgreSQL, AWS, Redis · Merged with Financial Modeling Prep

- Joined as the first engineering hire with no formal CS background; self-taught and shipped paid production features within months.
- Built the SEC Dashboard end-to-end: NLP classification pipeline (winkNLP, VADER), parsing SEC 8-K filings into summarized, investor-readable insights with keyword extraction and price-change context shipped as a full product feature.
- Integrated Stripe end-to-end, enabling Utradea's first paid users and establishing the monetization foundation.
- Piloted an early OpenAI API integration powering Financial Writer, an AI-generated news article tool built on real-time market data, shipped before ChatGPT made this obvious.
- Led post-merger migration of 50+ FMP APIs into Node.js/MongoDB; re-platformed Utradea from Lightsail onto ECS with Fargate and load balancers as traffic scaled, owned the full infrastructure transition.
- Implemented Redis cycle caching for high-frequency stock price reads with subscription-tier rate limiting across a multi-node cluster, designed for read/write throughput at scale.

Business Analyst | BMO Financial Group

May 2018 – Mar 2021

Capital markets & commercial banking · Requirements, systems analysis & data

Senior Business Analyst

Nov 2020 – Mar 2021

- Led the decommissioning of a vendor risk management platform, owning business requirements and migration to a streamlined in-house replacement
- Defined E2E requirements and drove cross-functional delivery for the Credit Flow for Commercial Banking transformation

Business Systems / Data Analyst

May 2018 – Nov 2020

- Owned E2E Client Portal Integration project, streamlining business banking onboarding from prospecting to signing across legacy systems via API
- Enhanced Online Banking for Business experience as part of the client portal integration, coordinating across multiple platform teams
- Introduced configurable credit compliance reporting templates, scaled across multiple lines of business to standardize regulatory workflows

EDUCATION

Bachelor of Commerce (B.Com.) | University of Toronto Rotman Commerce

2018

Major in Management · Minor in Economics

Venture Growth in Machine Learning Stream | Creative Destruction Lab (CDL)

2017 – 2018

Student Cohort 2018 · Worked alongside ML-native founders and operators; developed early conviction in AI-native product building

SKILLS

Engineering design	TypeScript · PostgreSQL · Redis · AWS (ECS, RDS) · API design · LLM orchestration · System
Product	0→1 product development · Requirements definition · Roadmapping · PMF thinking
Domain	Capital markets · Commercial banking · Risk & compliance · Fintech infrastructure